NATURAL LANGUAGE PROCESSING

Outline

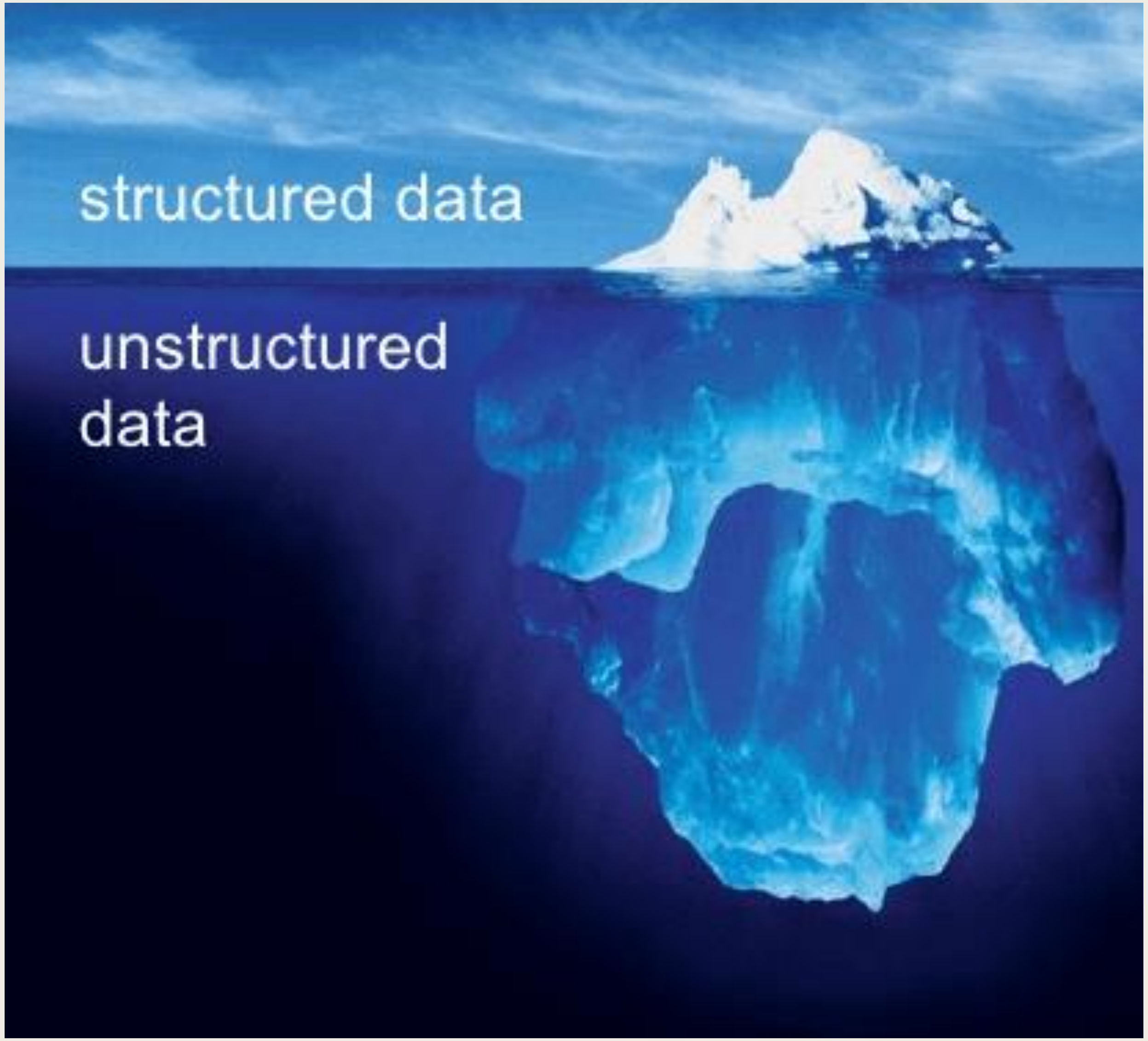
- Discuss natural language processing
- Examine elements of natural language processing pipeline including
  - *Text Acquisition*
  - *Text Cleaning*
  - *Tokenization*
  - *Normalization*
  - *Feature Extraction*
  - *Modeling & Evaluation*
  - *Sentiment Analysis*

Natural Language Processing…

- … is the field of computer science focused on enabling computers to understand, interpret, and generate human language.
- … bridges the gap between *human communication* and *machine understanding*
- *… involves linguistics, computer science, statistics, and machine learning*

Vast Majority of Data is Unstructured

But, even today, most decisions are based on structured data.



What is Unstructured Data?

- Data that does not fit into rows and columns in a spreadsheet such as
  - *Text*
  - *Pictures*
  - *Audio*
  - *Video*
- We will focus on Text

Text is everywhere

- Books and articles
- Web pages
- News stories
- Call center notes
- Doctor’s notes
- Emails
- Reviews
- Social media

Uses

- Business Intelligence: Sentiment analysis for brand monitoring.
- Healthcare: Extracting diagnoses from patient notes.
- Education: Automated essay scoring and feedback.
- Law: Contract summarization and clause extraction.
- Public Policy: Tracking opinions on social media.

Key Challenges

Analyzing text is hard

- Ambiguity: Words can have multiple meanings (e.g., bank, minute).
- Context Dependence: Meaning depends on surrounding words.
- Variability: Different ways to express the same idea.
- Noisy Data: Slang, typos, emojis, multilingual text (e.g., chinglish, hinglish, spanglish).
- Cultural Nuance: Language reflects culture and context, making universal interpretation hard.

NLP Pipeline

A structured process to transform raw text into meaningful data:

1. Text Acquisition — collecting documents, tweets, reviews
2. Text Cleaning — removing punctuation, symbols, stopwords
3. Tokenization — splitting text into words or sentences
4. Normalization — stemming, lemmatization, case folding
5. Feature Extraction — representing text numerically
6. Modeling & Evaluation — building ML models and assessing performance

Text Acquisition

Gather raw text data that represents the phenomenon of interest.

Sources of text:

- Web scraping (e.g., product reviews, news articles)
- APIs (Twitter, Reddit, PubMed, etc.)
- Internal documents, transcripts, or logs

But, recognize that data that is easily available may not be suitable

Text Cleaning

Remove irrelevant or noisy elements that hinder analysis.

Common text cleaning steps

- Convert to lower case for consistency
- Remove whitespace
- Replace abbreviations (e.g., AI with Artificial Intelligence)
- Remove punctuation, HTML tags, URLs, emojis
- Eliminate numbers (if irrelevant)
- Filter stopwords (e.g., the, and, is)
- Handling misspellings and encoding errors

Tokenization

Splitting text into smaller linguistic units such as words or sentences. Tokens are the atomic units on which most NLP algorithms operate.

Tokenization Methods

- Word Tokenization: “NLP is powerful” --> [“NLP”, “is”, “powerful”]
- Sentence Tokenization: Breaking documents into sentences

Challenges:

- Handling contractions and hyphenations (“don’t”, “self-driving”)
- Language-specific rules

Normalization

- Standardize tokens to reduce linguistic variation
- Techniques:
  - *Stemming: Reduces words to their root form (running -> run)*
  - *Lemmatization: Reduces words to their dictionary form (better -> good).* Lemmatization is more linguistically accurate but computationally expensive.
  - *Case Folding: Converts all words to lowercase*
  - *Removing Accents or Diacritics: “café” → “cafe”*
- Tradeoff
  - *Normalization reduces number of unique tokens but may result in loss of nuance and meaning. It can be computationally expensive.*

Feature Extraction

Convert text into numerical representations for computation.

- Bag of Words
  - *Examines words but ignores order. Two types:*
    - Term Frequency: Frequency of occurrence in a document
    - TF-IDF Weighted: Accounts for frequency of word and uniqueness to the document
- Advanced Approaches
  - *Word Embeddings: Word2Vec, GloVe — capture semantic similarity.*
  - *Contextual Embeddings: BERT, GPT — meaning depends on context.*
- We will not examine advanced approaches in this class because they require significant computational resources (GPUs, large memory), large datasets for effective tuning. Moreover, the skills learnt with bag of words are easily transferable to advanced approaches.

Modeling and Evaluation

- Build and assess predictive or descriptive models using extracted features
- Modeling Tasks
  - *Text classification*
  - *Prediction*
  - *Other advanced applications such as summarization and translation*

Sentiment Analysis

- Determining the sentiment or emotion expressed in a piece of text, which can be anything from a social media post to a detailed product review.
- Sentiment may be binary (e.g., thumbs up, thumbs down), trinary (e.g., thumbs up, neutral, thumbs down) or have multiple levels (e.g., very bad, bad, neutral, good, very good). Sentiment may also reflect emotions like joy, surprise or horror.
- Sentiment can be measured using
  - *Rules-based approach: Tokenized text is classified based on a sentiment lexicon.*
  - *Machine-Learning approach: Tokenized text are used as features to fit a model which is then used to score unseen data.*

Conclusion

- In this module, we
  - *Discussed natural language processing*
  - *Examine elements of natural language processing pipeline including*
    - Text Acquisition
    - Text Cleaning
    - Tokenization
    - Normalization
    - Feature Extraction
    - Modeling & Evaluation
    - Sentiment Analysis